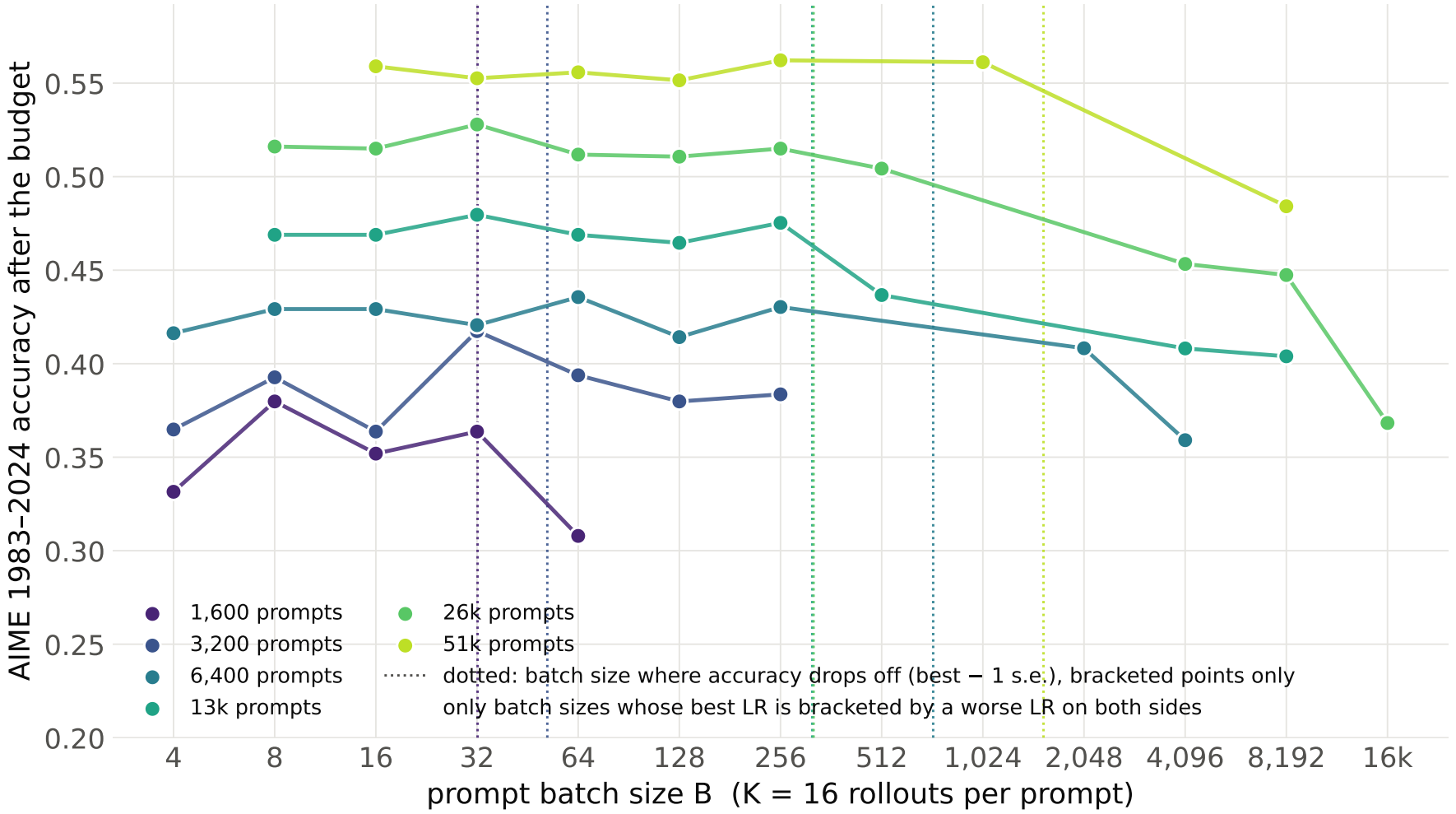
accuracy after a fixed number of prompts trained on, best learning rate per point, KL 0.001
AIME 1983–2024 accuracy after the budget
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
4
8
16
32
64
128
256
512
1,024
2,048
4,096
8,192
16k
prompt batch size B (K = 16 rollouts per prompt)
1,600 prompts
3,200 prompts
6,400 prompts
13k prompts
26k prompts
51k prompts
dotted: batch size where accuracy drops off (best − 1 s.e.), bracketed points only
only batch sizes whose best LR is bracketed by a worse LR on both sides